

Embedded Linux & System Software Interview Handbook

Phase 9 – Embedded Linux Build Systems & BSP Engineering

This phase covers the complete Embedded Linux development workflow, including build systems, cross-compilation, BSP engineering, boot image generation, and production deployment.

Chapter 16 – Embedded Linux Build Systems & BSP Engineering

Topics Covered

- Embedded Linux Architecture
- Board Support Package (BSP)
- Cross Compilation Toolchains
- GNU Toolchain (GCC, Binutils, GDB)
- Buildroot
- Yocto Project
- OpenEmbedded
- BitBake
- BusyBox
- Root Filesystem Creation
- Init Systems (SysVinit, systemd)
- Boot Image Generation
- Kernel Configuration (menuconfig)
- Device Tree Compilation
- Kernel Modules
- Initramfs and Initrd
- U-Boot Integration
- Flashing Images (eMMC, SD, NAND, NOR)
- Partition Layout
- OTA (Over-the-Air) Updates
- A/B Partitioning
- Package Management
- Firmware Updates
- Board Bring-up Workflow
- Boot Time Optimization
- Power Optimization
- Embedded Linux Debugging
- Production Deployment

- Manufacturing Considerations
- Real BSP Case Studies
- Senior Interview Questions

Learning Outcomes

After completing this phase, you will understand how to build a complete Embedded Linux distribution, create and customize BSPs, generate bootable images, optimize boot time, and deploy Linux on production hardware.

Phase	Coverage
Phase 1	Core Linux Kernel Fundamentals
Phase 2	Memory & Process Management
Phase 3	Filesystems & Storage
Phase 4	Linux Networking
Phase 5	Interrupts, Timers & Deferred Work
Phase 6	Advanced Device Drivers
Phase 7	Linux Security & Secure Boot
Phase 8	Performance & Debugging
Phase 9	Embedded Linux Build Systems & BSP
Estimated Size	250–350 pages when expanded
Target	Senior Embedded Linux / BSP / Platform Engineering Interviews